The law of the empirical frequency vector differs from the Dirichlet limit only off two exceptional events

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Editorial note. The counted statement is the author's labelled lem environment at source0.tex lines 1266-1272; its proof environment opens at line 1274, the line immediately after the blank separator, so the pairing is by adjacency and the retained proof range 1274-1298 is the whole environment.

Editorial note. Two environments are retained as context because the counted proof cites them: the author's coupling lemma (source lines 1159-1166), used at lines 1274 and 1295, and the numbered transition estimate eq:transition delta to delta* (source lines 1219-1222), used at line 1287.

Editorial note. This article carries no label-number dump, so no counter values are imposed: the retained environments print with the numbering the wrapper produces, and the identity of the counted item is its author label lem:sym diff D and Z.

Editorial note. The retained preamble is the publisher layout adapter, the native font and hyphenation packages and the native mathematical macro and theorem block of the pinned source; the article bibliography is bundled because the retained proof carries no citation of its own but the source does elsewhere in the same file.

Editorial note. r002 note: the two numbered equations eq:def of Theta_x (source lines 504-507) and eq:defn delta (source lines 1153-1156) are retained as context because the retained coupling-lemma context refers to them at source line 1165; r001 failed only on those two undefined references.

Editorial scope. One original result of the pinned publisher article, counted once: the comparison of the law of the empirical frequency vector with its Dirichlet limit off two exceptional events (author label lem:sym diff D and Z).

Allowed standard prerequisites (editorial).

- Empirical frequency vector and Dirichlet limit. delta_{f,x} is the vector of normalised logarithms of the ordered parts of a random factorisation, and Z is the Dirichlet limit vector.. They are the two random vectors whose laws the statement compares.
- The events E, E_cnt(A) and E_prob(A). E is the coupling event of the author's coupling lemma; E_cnt(A) and E_prob(A) are the events that

the two relevant closed balls avoid the boundary of the measurable set A.. The statement bounds the discrepancy by the complements of these events plus 1/log x, and the proof works on their intersection.

- Coupling distance bound. On the coupling event E one has a uniform bound on the maximal norm of the difference of delta_{f,x} and Z, in terms of log(x/N_x), log s(N_x) and Theta_x.. It supplies the radius within which a point of the connecting segment is exhibited.

$$\Theta_x \coloneqq \sum_{i\geq 1} r(V_i \log x). \tag{1}$$

$$\boldsymbol{\delta}_{f,x} \coloneqq \Big(\frac{\log D_{f,x,1}}{\log N_x}, \ldots, \frac{\log D_{f,x,k}}{\log N_x}\Big) \tag{2}$$

Lemma 0.1. *Let $x \geq 2$ and assume the above notation. There exists a coupling of N_x, $\boldsymbol{V}$, $\boldsymbol{D}_{f,x}$ and $\boldsymbol{Z}$ such that $\mathbb{P}[\mathcal{E}^c] \ll 1/\log x$ with*

$$\mathcal{E} \coloneqq \Big\{ \|\boldsymbol{\delta}_{f,x} - \boldsymbol{Z}\|_\infty \leq \frac{2\cdot\log(x/N_x) + 3\cdot\log s(N_x) + 2\cdot\Theta_x}{\log x}\Big\},$$

with $\boldsymbol{\delta}_{f,x}$ defined as in (2), Θ_x defined as in (1), and $\|\cdot\|_\infty$ being the supremum norm in $\mathbb{R}^k$.

$$\big\|\boldsymbol{\delta}_{f,x} - \boldsymbol{\delta}^*_{f,x}\big\|_\infty \leq \frac{\log(x/N_x)}{\log x}. \tag{3}$$

Lemma 0.2. *For any measurable set $A \subseteq \Delta^{k-1}$, we have*

$$\mathbb{P}\big[\boldsymbol{\delta}_{f,x} \in A\big] = \mathbb{P}\big[\boldsymbol{Z} \in A\big] + O\Big(\mathbb{P}\big[\mathcal{E}_{\mathrm{NT}}(A)^c\big] + \mathbb{P}\big[\mathcal{E}_{\mathrm{PD}}(A)^c\big] + \frac{1}{\log x}\Big).$$

Proof. Let $\mathcal{E}$ be as in Lemma 0.1. Hence, it suffices to show that

$$\Big|\mathbb{P}\big[\mathcal{E}\cap\{\boldsymbol{\delta}_{f,x}\in A\}\big] - \mathbb{P}\big[\mathcal{E}\cap\{\boldsymbol{Z}\in A\}\big]\Big| \leq \mathbb{P}\big[\mathcal{E}_{\mathrm{NT}}(A)^c\big] + \mathbb{P}\big[\mathcal{E}_{\mathrm{PD}}(A)^c\big]. \tag{4}$$

We shall prove the following stronger statement: if $\mathcal{E}\cap\mathcal{E}_{\mathrm{NT}}(A)\cap\mathcal{E}_{\mathrm{PD}}(A)$ occurs, then $\boldsymbol{\delta}_{f,x}\in A$ if, and only if, $\boldsymbol{Z}\in A$.

Indeed, assume that $\mathcal{E}\cap\mathcal{E}_{\mathrm{NT}}(A)\cap\mathcal{E}_{\mathrm{PD}}(A)$ occurs, and we also have $\boldsymbol{\delta}_{f,x}\in A$. Aiming for a contradiction, we also assume that $\boldsymbol{Z}\notin A$.

Let

$$R'_{\mathrm{NT}} \coloneqq (\log x)^{-1}(2\cdot\log(x/N_x) + 3\cdot\log s(N_x)).$$

Note that $B(\boldsymbol{\delta}_{f,x}, R'_{\mathrm{NT}}) \subseteq B(\boldsymbol{\delta}^*_{f,x}, R_{\mathrm{NT}})$ by (3). By assumption of $\mathcal{E}_{\mathrm{NT}}(A)$, we know that $B(\boldsymbol{\delta}_{f,x}, R'_{\mathrm{NT}})\cap\partial A = \emptyset$. Since $\boldsymbol{\delta}_{f,x}\in A$ and $B(\boldsymbol{\delta}_{f,x}, R'_{\mathrm{NT}})\cap\Delta^{k-1}$ is connected (being the intersection of two convex sets), we have that

$$B(\boldsymbol{\delta}_{f,x}, R'_{\mathrm{NT}})\cap\Delta^{k-1} \subseteq A.$$

And by assumption of $\mathcal{E}_{\mathrm{PD}}(A)$, we have $\boldsymbol{Z}\in\Delta^{k-1}\setminus A$ and $B(\boldsymbol{Z}, R_{\mathrm{PD}})\cap\partial A = \emptyset$, which implies

$$B(\boldsymbol{Z}, R_{\mathrm{PD}})\cap\Delta^{k-1} \subseteq \Delta^{k-1}\setminus A.$$

In particular, we find that $B(\boldsymbol{\delta}_{f,x}, R'_{\mathrm{NT}}) \cap B(\boldsymbol{Z}, R_{\mathrm{PD}}) \cap \Delta^{k-1} = \emptyset$. On the other hand, we know that $\|\boldsymbol{\delta}_{f,x} - \boldsymbol{Z}\|_\infty \leq R'_{\mathrm{NT}} + R_{\mathrm{PD}}$ when $\mathcal{E}$ occurs by Lemma 0.1. In particular, there exists a point $\boldsymbol{y}$ on the line segment connecting $\boldsymbol{\delta}_{f,x}$ and $\boldsymbol{Z}$ such that $\|\boldsymbol{\delta}_{f,x} - \boldsymbol{y}\|_\infty \leq R'_{\mathrm{NT}}$ and $\|\boldsymbol{y} - \boldsymbol{Z}\|_\infty \leq R_{\mathrm{PD}}$. Since the sets Δ^{k-1}, $B(\boldsymbol{\delta}_{f,x}, R'_{\mathrm{NT}})$ and $B(\boldsymbol{Z}, R_{\mathrm{PD}})$ are convex, we conclude that $\boldsymbol{y}$ lies in their intersection. But we had seen before that this intersection is the empty set. We have thus arrived at a contradiction. This completes the proof that if $\mathcal{E} \cap \mathcal{E}_{\mathrm{NT}}(A) \cap \mathcal{E}_{\mathrm{PD}}(A)$ occurs, and we also know that $\boldsymbol{\delta}_{f,x} \in A$, then we must also have that $\boldsymbol{Z} \in A$.

Conversely, we may show by a simple variation of the above argument that if $\mathcal{E} \cap \mathcal{E}_{\mathrm{NT}}(A) \cap \mathcal{E}_{\mathrm{PD}}(A)$ occurs, and we also know that $\boldsymbol{Z} \in A$, then we must also have that $\boldsymbol{\delta}_{f,x} \in A$. This completes the proof of the lemma. □

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